

Coarse to Fine Recursive Image-based Localization on a 3D Mesh Map

Yohei Hanaoka*
KDDI Research, Inc.

Kohei Matsuzaki†
KDDI Research, Inc.

Satoshi Komorita‡
KDDI Research, Inc.

ABSTRACT

Image-based localization is appropriate for augmented/virtual reality (AR/VR) services due to its high accuracy. However, it generally needs dedicated 3D map generation, which is costly. Thus, some methods have tried to utilize 3D mesh maps instead. However, the significant visual difference between the query image of the camera to be localized and a 3D mesh map results in low accuracy. This paper proposes to improve accuracy by dynamically controlling the comparison threshold and selecting the best localization result in a recursive estimation process. The results show that it improves the accuracy by 15% compared to the baseline method.

Index Terms: Visual Localization—Visual Positioning—3D Mesh Map

1 INTRODUCTION

The concept of a digital twin is spreading to create virtual spaces replicating real spaces and provide new experiences and values such as entertainment using AR/VR. The digital twin involves transferring information in real space to virtual space and projecting information in virtual space to real space through AR. An essential technology for realizing this concept is high-precision localization in real space to merge between virtual space and real space.

Image-based visual localization has the advantage of accurately determining pose including orientation and position, compared with radio wave-based methods like GPS and Bluetooth. On the other hand, this method generally needs to compare the query image of the camera to be localized with a dedicated 3D map containing image features. In practice, the creation of a 3D map is costly and laborious. Thus, some methods, *e.g.*, [4] have tried to localize by using a generic 3D mesh map, *e.g.*, PLATEAU [1] created for other purposes, such as VR services, instead. However, significant visual difference can be found between query photo and rendered image as shown in Fig. 1. In addition, the misalignment of the pose of the rendered image makes the images more visually different and more difficult to compare.

In this paper, we propose a coarse-to-fine method. In this scheme, coarse estimation is firstly employed considering the misalignment of rendered pose. And then, finer estimation is gradually employed as the misalignment is suppressed. In addition, our method selects the best results in the recursive process to improve the final accuracy.

2 RELATED WORK

In general, image-based visual localization using a 3D mesh map starts from rendering an image at a certain pose. The 2D correspondences of keypoints between the rendered image and the query image can be obtained by image comparison. Then, the 2D keypoints of the rendered image are back-projected to a 3D mesh map to get 3D coordinates. Given the correspondence between the 3D keypoints of the rendered image and the 2D keypoints of the query image, the

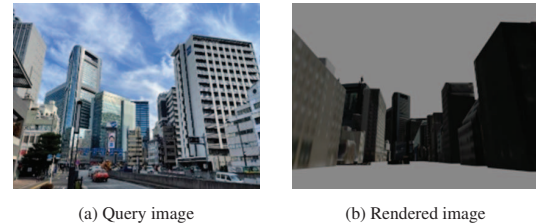


Figure 1: Major visual differences between query image and rendered image

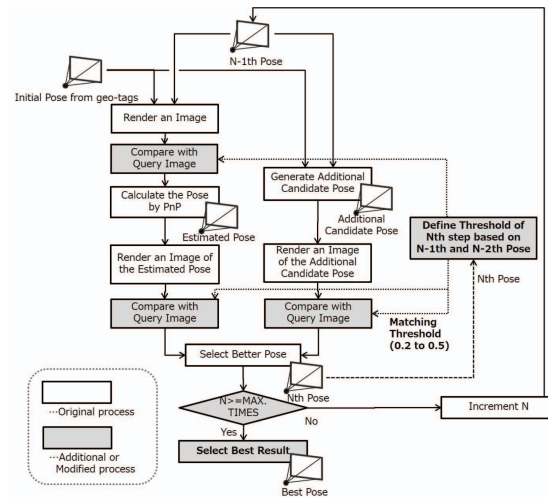


Figure 2: The proposed recursive procedure. Geo-tags are the data obtained from GPS and geomagnetic sensors. Gradually tighten comparisons according to the degree of convergence.

camera pose can be calculated by solving perspective-n-point (PnP). The image comparison is the essential part in establishing the 2D correspondences between the images. A minor decrease in visual similarity such as the illumination environment or viewpoints changes can lower the correspondences, reducing the pose accuracy. Thus, highly robust image comparison using deep-learning such as [3, 6] have been studied and improved. However, if the appearance is too different, as shown in Fig. 1, the accuracy is degraded.

MeshLoc [5] has been proposed recently, but it requires dense 3D meshes and does not work effectively with the rough textures and geometry of the generic 3D mesh map. We previously proposed the recursive method [4] that repeats rendering and comparison recursively for accurate localization, even if there are significant visual differences between the images. However, there are problems with not being able to start the recursive process due to initial sensor misalignment and the recursive process being terminated with insufficient accuracy.

3 PROPOSAL

Our proposal is based on the idea [4] that pose estimation by comparing the rendered image at a certain pose with the query image can give a more accurate pose, then it is repeated to obtain a more precise

*e-mail:yo-hanaoka@kddi.com

†e-mail:ko-matsuzaki@kddi.com

‡e-mail:sa-komorita@kddi.com

Table 1: Results of MLE, MAE and accuracy rate. The top two are in bold font and the best results are underlined

Methods	1m,2°(%)	2.5m,5°(%)	5m,10°(%)	MLE (m)	MAE (°)
Geo-tags	0.0	1.6	14.3	5.56	16.17
One-Time-Rendering	36.5	50.8	65.1	2.43	0.84
Prev (original) [4]	41.3	60.3	66.7	1.31	0.51
Prev + SuperGlue *	54.0	76.2	81.0	0.89	0.42
Ours with DTC	55.6	85.7	88.9	0.90	0.36
Ours with BPS	60.3	76.2	82.5	0.89	0.43
Ours with DTC and BPS	60.3	87.3	92.1	0.85	0.36

* A baseline method for our proposal

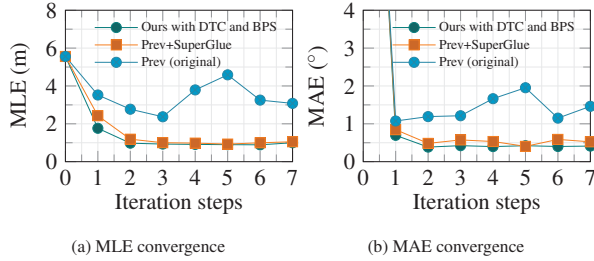


Figure 3: Comparison of error convergence. Initial MAE is 16.17°

location. During this repeated process, we apply the method that dynamically controls the strictness of the comparisons depending on the difficulty of the accurate comparison. In addition, we select the best results at the end of the process. The following provides the details of these proposals.

Dynamic Threshold Control (DTC) ensures appropriate image comparisons according to pose estimation accuracy. The threshold of the image comparisons is lowered in the initial comparison stage to increase the number of 2D keypoints and to obtain coarse comparative results. The threshold is raised for more rigorous and fine comparisons when accuracy improves. We use the difference in the position of the pose at each step as the estimation accuracy because the higher accuracy makes the estimated position stable. Fig. 2 shows the proposed flow of the recursive process combined with this method. The threshold used for comparison with the query image in the N th estimation process is determined from the difference between the N -1th and N -2th pose. Note that if $N < 2$, the default lower threshold is used.

Best Pose Selection (BPS) selects the most accurate pose after recursion up to the maximum number of times as shown in Fig. 2. We also use the difference between the estimated position and its previous one in this determination. The pose with the smallest difference is selected as the best pose.

4 EXPERIMENTS

We evaluated our proposal using a Tokyo-localization-2022 dataset [2]. For image comparison, we used the SuperPoint [3] and SuperGlue [6] methods that demonstrated the best performance in our preliminary experiments. The dynamic thresholds of DTC were experimentally set to 0.5, 0.4, 0.3, and 0.2 for the position difference of less than 1m, 1m to 2.5m, 2.5m to 5.0m, and more than 5m, respectively. Except for DTC, we used a fixed threshold of 0.5. The maximum numbers of iterations of the recursive process was set to 10. If BPS is not applied, the best result is the pose at which the distance from its previous estimated position is less than 1 m, according to the original settings. We evaluated our proposed method on three patterns: with DTC, with BPS, and with DTC and BPS. We also performed reference experiments using SuperGlue for comparison. "Geo-tags" is just the initial pose by GPS and magnetometer sensors. "One-Time-Rendering" uses only the initial rendered image

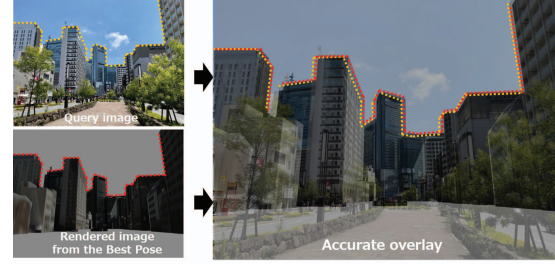


Figure 4: An example of the query image, the rendered image from the best pose, and their semi-transparent overlaid image. The red and yellow lines are the outlines of the building edges. It can be seen that the appearance of the images overlap approximately exactly. The localization error is 0.56m and angular error is 0.18°.

as a simple method in a renderable environment. Our previously proposed method [4], denoted as "Prev", was implemented using SuperGlue as a baseline for our proposal, in addition to the original implementation using LoFTR [7].

Table 1 shows the experimental results. The evaluation metrics are localization errors (LE)(m) and angular errors (AE)(°) between the estimated pose and the ground truth (GT). The accuracy rate (%) is the ratio of trials estimated within three error thresholds (1m, 2°), (2.5m, 5°), and (5m, 10°). MLE and MAE are the median of LE and AE, respectively. The results show that "Ours with DTC and BPS" has the best performance in terms of the accuracy rate and the mean of errors. In particular, accuracy below 2.5 m is 15% better than the baseline and 45% better than the original. Improved accuracy will enhance the experience value of location-based services such as AR. It also indicates that BPS is the most in improving accuracy at less than 1 m, while DTC improves accuracy at 1 m or more. Fig. 3 shows the convergence of each error. Although the convergence is greatly accelerated by SuperGlue, our proposal has even smaller errors approximately across the entire range.

Fig. 4 shows an example of the query image, the rendered image at the best pose, and their overlaid image. The closer the best pose is to the GT, the more closely the images overlap. You can see that the overlap is accurate and sufficient to overlay the approximate virtual information on real space in AR although the original method [4] had the boundary between the buildings misaligned.

5 CONCLUSION

We proposed a method to achieve more accurate pose estimation on a 3D mesh map. We confirmed that the proposal increased the accuracy ratio by 15% at 2.5 m or less and archived the best performance in terms of accuracy and errors.

REFERENCES

- [1] PLATEAU by mlit. <https://www.mlit.go.jp/plateau>, 2020.
- [2] Tokyo-localization-2022 dataset. <https://github.com/yohanaoka/tokyo-localization-2022>, 2022.
- [3] D. DeTone, T. Malisiewicz, and A. Rabinovich. Superpoint: Self-supervised interest point detection and description. In *CVPR Deep Learning for Visual SLAM Workshop*, 2018.
- [4] Y. Hanaoka, S. Suwanwimolkul, and S. Komorita. Recursive Rendering of 2D Images for Accurate Pose Estimation in a 3D Mesh Map. In *SIGGRAPH ASIA*, 2022.
- [5] V. Panek, Z. Kukulova, and T. Sattler. Meshloc: Mesh-based visual localization. In *European Conference on Computer Vision*, pp. 589–609. Springer, 2022.
- [6] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich. SuperGlue: Learning feature matching with graph neural networks. In *CVPR*, 2020.
- [7] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou. Loft: Detector-free local feature matching with transformers. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8922–8931, 2021.